

ServiceHub: An Android-Based Local Service Marketplace for Connecting Users with Nearby Service Providers

Abstract

ServiceHub is an Android-based mobile application designed to bridge the gap between customers and local service providers by offering a centralized platform for service discovery, booking, and management. In today's fast-paced environment, users often face difficulties in finding reliable professionals such as electricians, plumbers, carpenters, tutors, cleaners, painters, and other local service providers. Traditional methods of searching through contacts, advertisements, or local references are often time-consuming and unreliable. ServiceHub addresses this problem by providing a digital marketplace where customers can easily discover, compare, and book nearby service providers based on their requirements.

The application is developed using XML for the user interface design and Java for application logic and backend processing within the Android ecosystem. The system supports multiple user roles, including customers and service providers. Customers can create accounts, browse service categories, search for providers, view provider profiles, check ratings and reviews, book appointments, and track their service requests. Service providers can register their services, manage their profiles, update availability, receive booking requests, and maintain communication with customers through the application.

The application incorporates location-based services using GPS to help users identify nearby providers efficiently. A rating and review system promotes transparency and service quality. Authentication mechanisms ensure secure access to user accounts, while the booking management module enables scheduling and tracking of service requests.

The proposed system digitizes local service discovery and appointment management, enhancing convenience, accessibility, and operational efficiency. It demonstrates Android development concepts such as Activities, Intents, RecyclerView, authentication, database management, GPS integration, and real-time data handling.

Objectives

- Provide a platform for discovering local service providers.
- Simplify service booking through a mobile application.
- Enable location-based search for nearby professionals.
- Implement a secure user authentication system.
- Provide ratings and reviews for service quality assessment.
- Improve communication between customers and service providers.
- Digitize appointment scheduling and booking management.

Modules

- Customer Module – Registration, Login, Search, Booking, Reviews.
- Service Provider Module – Profile, Services, Availability, Booking Management.
- Location Module – GPS Access, Nearby Service Discovery, Distance Calculation.

- Booking Module – Scheduling, Status Tracking, Service History.
- Review Module – Ratings, Feedback, Reputation Management.

Technologies Used

- Frontend: XML, Material Design Components
- Backend Logic: Java
- Database: SQLite / Firebase
- Development Environment: Android Studio
- Platform: Android OS

Expected Outcome

- Provide an efficient platform for finding, booking, and reviewing local service providers.
- Reduce service discovery time and improve booking efficiency.
- Offer professionals a digital platform to expand their customer reach.